Proposal: Do LLM Rankings Change if You Ask Differently? Evaluating LLM Ranking Reversals Under Prompt Perturbations

Group ID: 14

Administrative Information

Group members	ZHANG QIANCHI (A0333967N), PENG YANGYUNZHI (A0333686U), JIANG YIFAN (A0333201B), MENG XIANGCHEN (A0327078X)
Chosen theme	**(check one)**: [✓] Reliability of LLM Evaluation [☐] Validity of LLM-Based Judgments and Preferences [☐] Behavioral Variability Under Implicit Conditions [☐] Decision Consistency and Threshold Effects
Compute plan	[✓] API-based [☐] open-source [☐] mixed
Reproducibility plan	We will release all prompts, sampled question IDs, random seeds, and evaluation scripts. Model IDs and decoding parameters will be logged to ensure the evaluation pipeline can be reproduced.

Abstract

Large language model (LLM) leaderboards are widely used to compare models, yet it is unclear whether these rankings are stable under small evaluation changes. In this project, we study whether non-semantic prompt variations and sampling noise can significantly affect reported model performance and rankings. We evaluate four API-accessible models via OpenRouter on three multiple-choice benchmarks (MMLU, ARC-Challenge, and HellaSwag). For each question, we test several prompt templates that differ only in wording or formatting. We further estimate evaluation uncertainty using bootstrap-style resampling of the evaluation set and recompute model accuracies and rankings. Our goal is to quantify how often small evaluation changes lead to score shifts or ranking reversals.

1. Research Question

In this study, we investigate whether small variation in prompt and evaluation sets significantly affect reported LLM performance scores, and further, whether these fluctuations are sufficient to trigger ranking reversals among top-tier models, rendering standard leaderboards unreliable.

2. Motivation and Related Work

- **Instability & Risk:** LLM rankings exhibit extreme sensitivity to both **prompting choices** and **data composition**. Non-semantic perturbations (e.g., phrasing, formatting) and minor fluctuations in **test subsets** can trigger accuracy changes that exceed the performance gap between top models (Razavi et al., 2025; Miller, 2024). This suggests that rankings often stem from "surface-form luck" or "sampling noise" rather than robust logical superiority (Min et al., 2022), posing a practical risk for practitioners who rely on unstable point-estimates to select models.
- **Importance for:** *Benchmark Designers* must move beyond point-estimates to maintain credibility (Fourrier et al., 2024); *Researchers* require rigorous statistical protocols to distinguish genuine reasoning improvements from stochastic noise (Miller, 2024).
- **Our work:** While prior work catalogs score variance under prompt changes (Razavi et al., 2025), we focus on the **structural stability of relative rankings**. By integrating bootstrap resampling with prompt perturbations, we analyze how model hierarchies collapse under the joint noise of **formatting and data sampling**. This provides a multi-dimensional reliability signal—covering both "prompt luck" and "test-set noise"—that is currently absent in point-estimate leaderboards.

3. Experimental Setup

3.1 Models

We will conduct evaluations on the following API-accessible chat models through OpenRouter:

1. OpenAI: GPT-4.1 Mini (`openai/gpt-4.1-mini`)
2. Anthropic: Claude 3.5 Haiku (`anthropic/claude-3.5-haiku`)
3. Google: Gemini 2.0 Flash (`google/gemini-2.0-flash-001`)
4. Meta: Llama 3.1 8B Instruct (`meta-llama/llama-3.1-8b-instruct`)

3.2 Datasets

We will use three automatically scorable multiple-choice benchmarks:

Dataset/Task	**Size**	**Sampling / split / inclusion criteria**
MMLU	300 items	Sampled from the official evaluation split, stratified across subjects where possible
ARC-Challenge	300 items	Sampled from the official evaluation split; only standard multiple-choice items retained
HellaSwag	300 items	Sampled from the validation/evaluation split; standard 4-option completion format

3.3 Evaluation Settings

We will experiment on evaluation settings designed from the two following perspectives:

1. Prompt variations: For each item, we use four task-preserving prompt templates:
   (a) minimal instruction: direct zero-shot prompt with "Answer with one option only."
   (b) benchmark-style instruction: a slightly more formal version with clearer task framing.
   (c) minor non-semantic variations: Adding minor perturbations such as different punctuations into the question, while preserving equivalence semantically.
   (d) order/phrasing variation: semantically equivalent instruction with different ordering of context and answer-format reminder.
2. Subsets: For each dataset, we first test models on a fixed master sample of $n = 300$ questions, then on $r = 100$ resampled subsets at the evaluation-table level from the master sample.

4. Metrics and Evaluation Protocol

4.1 Primary metrics

For each setting, we measure model performance by **accuracy**, i.e. the proportion of correctly answered questions. We further consider the **Ranking Reversal Rate (RRR)** among evaluations. For each dataset and a set of models, its RRR is defined to be the rate of occurrence of any model rankings different from the resulting ranking from master sample with minimal prompt.

4.2 Statistical protocol

For each dataset, we will use the resulting model accuracies on $r = 100$ subsets to get empirical distribution, report mean accuracy and 95% bootstrap confidence intervals. For model comparisons, we compute paired accuracy differences on each subset to estimate the uncertainty of reported performance gaps. We summarize the distribution of ranks and RRR to assess whether observed model differences remain stable under resampling and prompt variation.

5. Analysis Plan

Our comparison on model accuracy across prompt templates and resamples will be summarized with boxplots, 95% bootstrap confidence intervals, and pairwise ranking heatmaps. Our goal is to test whether small prompt variations and subset changes can alter leaderboard rankings.

6. Expected Challenges, Limitations, and Ethics

Considering API cost and rate limits, we keep the evaluation scale controlled. Due to contamination concerns on public benchmarks, we only focus on evaluation stability. Our results are limited to multiple-choice tasks. We only use public benchmark data and no personal or sensitive information.

References

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